Research Paper Review

Scaling Blockchain Consensus via a Robust Shared Mempool

Blockchain Technology & Applications
Name: Namrata Paralkar | PRN: 202402100007 | Class: Software Engineering

Literature search: three papers referred

Paper	Type / year	Focus
Scaling Blockchain Consensus via a Robust Shared Mempool (Gai et al.)	Protocol, arXiv 2022	Removes leader bottleneck in BFT consensus
A Miniscule Survey on Blockchain Scalability (Li, Song, Zhu)	Survey, arXiv 2022	Scalability, security, healthcare / supply chain / government uses
Performance Modeling of Public Permissionless Blockchains (Esmaili, Christensen)	Survey, arXiv 2024	Queueing-based performance models, workload modelling

Selected: Paper 1. It proposes a technique, is tested experimentally, and links to consensus and scalability topics in the syllabus.

(a) Paper details

- Title: Scaling Blockchain Consensus via a Robust Shared Mempool
- Authors: Fangyu Gai, Jianyu Niu, Ivan Beschastnikh, Chen Feng, Sheng Wang
- Venue: arXiv:2203.05158 [cs.DC]
- Submitted 10 Mar 2022; revised (v3) 25 Sep 2022
- DOI: 10.48550/arXiv.2203.05158

(b) Research problem

- Permissioned blockchains often use leader-based BFT consensus (PBFT, HotStuff-style)
- The leader distributes all transaction data and coordinates agreement
- With hundreds of replicas, leader bandwidth and CPU become the bottleneck
- Result: limited scalability and poor robustness under attack or slow networks
- Question: how to scale BFT without the leader limiting the system?

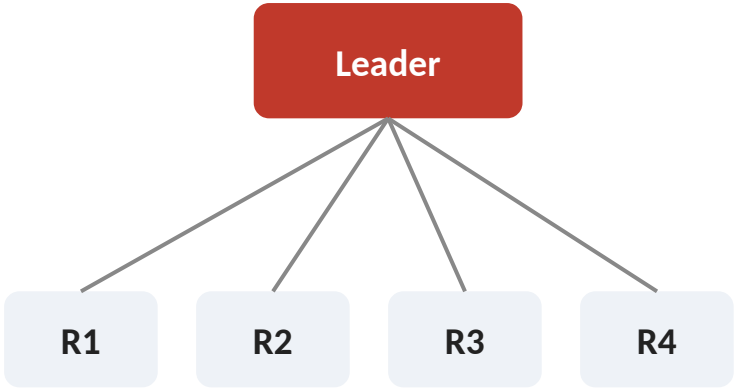
(c) Methodology: Stratus

- Shared-mempool layer that decouples transaction distribution from consensus
- Every replica disseminates its own transaction batches
- The leader proposes only transaction IDs
- Provably Available Broadcast (PAB) guarantees referenced transactions can be retrieved
- Load balancing for uneven workloads; simple APIs for other BFT systems

Evaluation: integrated with state-of-the-art BFT protocols; LAN and WAN tests with hundreds of replicas; asynchrony, Byzantine attackers, unbalanced load.

Idea in one picture: who carries the data?

Native leader-based BFT



Leader sends everything to everyone

With Stratus



Replicas share batches with each other, proven available by PAB

(d) Contributions and findings

- Modular shared-mempool protocol that removes the leader bottleneck
- PAB: guarantees availability of referenced transactions
- Graceful degradation under asynchrony, Byzantine attackers, unbalanced load
- 5 to 20 times higher throughput than native protocols with hundreds of replicas
- Reusable API for other BFT systems

(e) Limitations and future work

- Scope: permissioned, leader-based BFT; not directly applicable to PoW/PoS public chains
- Results from controlled LAN/WAN testbeds, not production networks
- Availability proofs add messages and crypto cost; latency at low load needs study
- Adaptive or collusive attacks on load balancing are not fully explored
- Future: combine with sharding or layer-2, dynamic membership, Hyperledger-style deployment

Limitations are my own critical assessment.

(f) Personal reflection

- Connects consensus (PBFT), permissioned vs public chains, and the scalability trilemma
- Shows a blockchain is a stack of parts that can be optimised independently
- Fits consortium uses: supply chain, interbank settlement, health records, government registries
- Gain comes from protocol design, not just bigger hardware
- Next step: reproduce a small-scale experiment

References

[1] F. Gai, J. Niu, I. Beschastnikh, C. Feng, S. Wang, "Scaling Blockchain Consensus via a Robust Shared Mempool," arXiv:2203.05158, 2022.

[2] A. Li, G. Song, T. Zhu, "A Miniscule Survey on Blockchain Scalability," arXiv:2212.13353, 2022.

[3] M. Esmaili, K. Christensen, "Performance Modeling of Public Permissionless Blockchains: A Survey," arXiv:2402.18049, 2024.